

The p^+ –integrated four–kernel row

Is it right, where the rapidity logarithm is, and what comes next

$$\kappa \equiv k \cdot (y' - z), \quad \xi \equiv p^+ / k^+, \quad \lambda \equiv \Lambda / k^+, \quad \Xi \equiv (\vee - k^+) / k^+, \quad \ell_k \equiv \log(k^+ / \Lambda) = -\log \lambda$$

1. Verdict

All three p^+ integrals are correct — structures, signs, arguments, and the $e^{ik \cdot (y' - z)}$ prefactor on the third line. Each was checked against numerical quadrature to 10^{-29} relative.

One thing to check: the overall $1/k^+$. See §2.

The bracket before integration is $\frac{\delta_{k'm} \delta_{ij}}{p^+ + k^+} - \frac{\delta_{im} \delta_{jk'}}{p^+} - \frac{\delta_{im} \delta_{jk'}}{k^+}$, which on the four kernels collapses to

$$\frac{1}{k^+} \left[\frac{A}{1 + \xi} - \frac{B}{\xi} - C \right], \quad \begin{aligned} A &= (y' - z) \cdot (x' - y') (y - w) \cdot (x - z) \\ B &= (y' - z) \cdot (x - z) (x' - y') \cdot (y - w) \\ C &= (y' - z) \cdot (y - w) (x' - y') \cdot (x - z) \end{aligned}$$

(all divided by $(y' - z)^2 (x' - y')^2 (y - w)^2 (x - z)^2$), with the phase $e^{-i\kappa\xi}$ from $e^{-ip \cdot (y' - z)}$ at $p_\perp = \xi k$. Then $\frac{1}{k^+} \int dp^+ = \int d\xi$ and

structure	$\int_\lambda^\Xi d\xi (\cdot) e^{-i\kappa\xi}$	your line	check
$-C$	$\frac{e^{-i\kappa\Xi} - e^{-i\kappa\lambda}}{i\kappa}$	1st	4.8×10^{-29}
$-B/\xi$	$-[\text{Ei}(-i\kappa\Xi) - \text{Ei}(-i\kappa\lambda)]$	2nd	1.3×10^{-30}
$+A/(1 + \xi)$	$e^{i\kappa} [\text{Ei}(-i\kappa(1 + \Xi)) - \text{Ei}(-i\kappa(1 + \lambda))]$	3rd	5.3×10^{-31}

Every one of these is exactly what you wrote, including the shift $1 + \Xi = \vee / k^+$ inside the last Ei and the $e^{ik \cdot (y' - z)}$ that the shift $u = 1 + \xi$ generates.

2. The overall $1/k^+$ — correct as written

An earlier draft of this note flagged the overall $\frac{1}{k^+}$ as possibly counted twice. **That flag was wrong;** the $\frac{1}{k^+}$ is right, for two independent reasons.

(i) It is not the $\frac{1}{k^+}$ from the bracket. The w' delta–function sets $w' = y' + \frac{p^+}{k^+}(y' - z)$, so $w' - z = \frac{p^+ + k^+}{k^+}(y' - z)$ and the first kernel supplies $\frac{k^+}{p^+ + k^+}$. Against the $\frac{1}{k^+}$ in front, and pulling $(p^+ + k^+)$ out of $[\delta_{k'm} \delta_{ij} - \frac{p^+ + k^+}{p^+} \delta_{jm} \delta_{ik'} - \frac{p^+ + k^+}{k^+} \delta_{im} \delta_{jk'}]$, this gives exactly $\frac{1}{k^+} [\frac{\delta_{k'm} \delta_{ij}}{p^+ + k^+} - \frac{\delta_{jm} \delta_{ik'}}{p^+} - \frac{\delta_{im} \delta_{jk'}}{k^+}]$ (checked symbolically: the difference is identically zero).

(ii) The measure does not eat it. Every term of that bracket carries exactly one inverse power of momentum, so with $p^+ = k^+ \xi$ and $dp^+ = k^+ d\xi$,

$$\int dp^+ \frac{1}{p^+ + k^+} = \int \frac{d\xi}{1 + \xi}, \quad \int dp^+ \frac{1}{p^+} = \int \frac{d\xi}{\xi}, \quad \int dp^+ \frac{1}{k^+} = \int d\xi,$$

the k^+ cancelling *inside* the bracket each time. So $\int dp^+ [\text{bracket}] = \int d\xi [\frac{A}{1 + \xi} - \frac{B}{\xi} - C]$ and the prefactor $\frac{1}{k^+}$ survives untouched. The denominator $i k \cdot (y' - z)$ in the first line is then also correct as it stands.

3. Extracting the rapidity logarithm

Two expansions do all the work. For $\varepsilon \rightarrow 0^+$,

$$\text{Ei}(-i\kappa\varepsilon) = \gamma_E + \log(i\kappa\varepsilon) + O(\kappa\varepsilon), \quad \text{Ei}(-i\kappa X) \xrightarrow{X \rightarrow \infty} -i\pi \operatorname{sgn} \kappa.$$

The second is worth pausing on: Ei at large imaginary argument does **not** go to zero, it goes to the constant $\mp i\pi$ ($|\text{Ei}(-i 10^6)| = 3.141592$). Dropping the upper limit is therefore wrong by $i\pi$.

Line by line, as $\Lambda \rightarrow 0$ and $\vee \rightarrow \infty$

Third line, $A/(1+\xi)$: no rapidity log. The denominator $1+\xi$ is regular at $\xi=0$, so $\text{Ei}(-i\kappa(1+\lambda)) \rightarrow \text{Ei}(-i\kappa)$ smoothly:

$$\longrightarrow e^{i\kappa} [-i\pi \text{sgn } \kappa - \text{Ei}(-i\kappa)].$$

First line, $-C$: no rapidity log. $e^{-i\kappa\lambda} \rightarrow 1$, and the Ξ piece is bounded and oscillatory (Riemann–Lebesgue kills it under the transverse integration):

$$\longrightarrow \frac{i}{\kappa} + (\text{oscillatory}).$$

This also retires the worry I raised last time that this term looked linear in $\vee$. It is not — the phase saves it.

Second line, $-B/\xi$: the sole rapidity divergence. In the double limit the bracket collapses to a single logarithm,

$$-[\text{Ei}(-i\kappa\Xi) - \text{Ei}(-i\kappa\lambda)] \longrightarrow \gamma_E + \log(i\kappa\lambda) = \boxed{-\ell_k + \gamma_E + \log(i k \cdot (y' - z))}$$

verified to 10^{-9} for both signs of κ (the log is the principal branch, and it carries the $\pm i\pi/2$ automatically). **The coefficient of ℓ_k is -1 , and it sits on the structure B alone.**

4. What multiplies ℓ_k is the BK kernel

Strip the ℓ_k off and look at what it multiplies:

$$-\ell_k \underbrace{\frac{(x-z) \cdot (y'-z)}{(x-z)^2 (y'-z)^2}}_{\mathcal{K}(x, y'; z)} \times \underbrace{\frac{(x'-y') \cdot (y-w)}{(x'-y')^2 (y-w)^2}}_{\text{no } z: \text{ LO structure}}$$

The first factor is *exactly* the real-emission piece of the Balitsky–Kovchegov / JIMWLK dipole kernel, with z the emission point and x, y' the two legs. That is the strongest available check that the p^+ integration is right: the rapidity divergence had to come out proportional to the evolution kernel, and it does.

And this is where last note's transverse IR log goes

In the previous note I found a logarithmic divergence of the z -integration at large $|z|$, of coefficient $-\pi\delta^{mj}$, and observed that it is tied to $\xi \rightarrow 0$. The full BK kernel is

$$\mathcal{M}(x, y'; z) = \frac{(x-y')^2}{(x-z)^2 (y'-z)^2} = \frac{1}{(x-z)^2} + \frac{1}{(y'-z)^2} - 2\mathcal{K}(x, y'; z),$$

and the two $1/(\cdot)^2$ terms are the **virtual** (self-energy / same-side) contributions of the same order, which are *not* in the row you wrote. Their large- $|z|$ angular averages are what cancel the divergence:

ρ	$\rho^2 \langle \frac{1}{(x-z)^2} \rangle$	$\rho^2 \langle \frac{1}{(y'-z)^2} \rangle$	$\rho^2 \langle -2\mathcal{K} \rangle$	$\rho^2 \langle \mathcal{M} \rangle$
10^1	1.001302	1.008980	-1.993815	0.016466
10^2	1.000013	1.000089	-1.999938	0.000164
10^3	1.000000	1.000001	-1.999999	0.000002
10^4	1.000000	1.000000	-2.000000	0.000000

$1 + 1 - 2 = 0$: the full kernel falls as ρ^{-4} , not ρ^{-2} , and the z -integral converges. **So the transverse infrared logarithm is not a disease of your calculation — it is telling you that the virtual diagrams are still missing.**

5. What to do next

1. **Settle the $1/k^+$ (§2)** against the pre-integration expression.
2. **Take $\vee \rightarrow \infty$ honestly.** Keep the $-i\pi \text{sgn } \kappa$ from $\text{Ei}(-i\kappa\Xi)$ — it is a finite constant, not zero, and it contributes to the imaginary (absorptive) part.

3. **Write the three brackets in their limiting forms** (§3). You then have a clean split: one term $\propto \ell_k$, everything else ℓ_k -independent.
4. **Collect the virtual partners** so that $-2\mathcal{K} \rightarrow \mathcal{M}$. Until you do, the z -integral is IR divergent and no separation into “finite” pieces is meaningful.
5. **Reduce the colour structure.** $f^{c'd'a}$ contracted with $[UU\rho - UU\rho][UUU\rho\rho - U\rho\rho]$ has to be brought to dipole / quadrupole operators before the ℓ_k coefficient can be compared with $\mathcal{H}_{\text{JIMWLK}}$. The check to aim for is

$$\left. \frac{d^3 N_{\text{NLO}}}{d^3 k} \right|_{\ell_k} = -\ell_k \mathcal{H}_{\text{JIMWLK}} \otimes \frac{d^3 N_{\text{LO}}}{d^3 k}.$$

6. **Absorb, don't cancel.** Once (5) holds, define the target weight functional at rapidity $Y = \ell_k$ and subtract; what remains is the genuine NLO impact factor.
7. **Then** do the transverse integrals with the master $B^{mj}(Q, c)$ of the previous note — but note the phase is now gone (you integrated p^+ first), so the roles of Λ and Q are swapped: Λ is what cuts the large-separation region, and the leftover $\log(i k \cdot (y' - z))$ from §3 is what turns the single ℓ_k into the double $\log \ell_k^2$ found earlier.

`zint/src/pplus_check.py`, `zint/src/pplus_limits.py` — every number above, written to `zint/RESULTS_pplus.txt`.